AUCTUM LEDGER — Document Scanner & Trust Score

Consolidated Team Review and Final Sprint Plan

Prepared by: Architecture / Systems Design / Tech Debt / Documentation / Design System / Design Handoff / Design Critique / Process Optimization / Process Documentation **Sources reviewed:** `Document_Scanner_Specification (Auctum Ledger)`, `Auctum_Ledger_Document_Scanner_Planning` (byte-identical duplicate), `Trust_Score_and_OCR_UI_Specification`, `Coffee_Industry_Omni-Expert`, `Experts_descriptions` **Prior context:** Greensheet brand positions, seed-to-cup verification workflow, exception-handling tiers, retained-sample SOP, cupping SOP, supplier agreement **Status:** Draft for founder sign-off. Nine ADRs require a decision before Sprint 1 opens.

PART 0 — READ FIRST

0.1 Credential exposure (blocking, do today)

`Experts_descriptions__1_.docx` contains what appear to be live API keys: two Anthropic `sk-ant-api03-*` keys, three other `sk-*` provider keys, and one hex-format key. This was flagged in a previous session and remains unremediated.

Action: revoke and rotate every key in that document, audit provider usage logs for unauthorized calls, remove the credentials from the document and from any Drive copy, and re-derive the role prompts into a clean file. No sprint work should begin against a repo or document set that carries live secrets.

0.2 Documentation debt, minor but telling

`Auctum_Ledger_Document_Scanner_Planning.docx` and `Document_Scanner_SpecificationAuctum_Ledger_.docx` are the same file (identical MD5). Two names for one artifact is how a spec set starts to drift. Pick one canonical name, delete the other, and put the spec in the repo under `docs/` where it versions with the code rather than in Drive where it forks.

PART 1 — TEAM REVIEW OF THE EXISTING SPECS

1.1 What the two specs actually are

They are not one plan in two parts. They are two plans written at different moments against different assumptions about what already exists.

	Document Scanner Spec	**Trust Score Spec**
Assumed starting point	Nothing built; port `marcusdax/vision-ocr-ai-agent-logistics`	`services/docintake` **already live** in `marcusdax/greensheet`
OCR architecture	Multi-engine (cloud vision + Tesseract + layout model) → separate LLM extraction step	Claude vision with structured outputs, single pass
Product name	Auctum Ledger	Auctum Ledger (formerly Greensheet)
Design system	Generic ("confidence badges green/amber/red")	Museum Folio tokens, specified in detail
Implementation sequence	10 steps	9 steps
Confidence thresholds	Doc-level ≥0.92, critical field ≥0.85	Field-level green ≥0.85, amber 0.70–0.84

Both are good documents. Read together they contain three architectures for one subsystem, two threshold schemes, and two sequencing orders that contradict each other on when the Review UI gets built relative to the confidence logic.

The first task of Sprint 0 is not building. It is establishing which of these describes reality.

1.2 Conflict register

ID	Conflict	Severity	Resolution
C-01	Is `docintake` live or does OCR need to be built?	🔴 Critical	Sprint 0 audit. Everything downstream branches here. See ADR-002.
C-02	Multi-engine OCR + LLM extraction vs. single vision-LLM	🔴 Critical	Single vision-LLM primary; deterministic second engine only as cross-check on money and score fields. ADR-002.
C-03	Threshold schemes differ (0.92/0.85 doc-level vs 0.85/0.70 field-level)	🟡 High	One versioned `confidence-policy` module in `contracts/`, consumed by worker and UI. ADR-004.
C-04	`documents` is tenant-scoped; Trust "travels with the profile" across tenants	🔴 Critical	Two-layer model: platform `parties` registry + tenant-private document view. ADR-001.
C-05	"Convert all money to integer cents" vs VND, which has no minor unit	🟡 High	Per-currency scale + FX record. ADR-003.
C-06	Repo/package/table names say `greensheet`, product says Auctum	🟡 High	Rename now at package and event level, or write down the boundary and live with it. ADR-008.

ID	Conflict	Severity	Resolution
C-07	"Store last 30 snapshots (or Timescale hypertable)" — undecided stack	🟢 Moderate	Append-only Postgres table. No Timescale in v1. ADR-005.
C-08	Scanner spec builds Review UI (step 6) before thresholds (step 8)	🟢 Moderate	Thresholds precede the UI that renders them. Corrected in Part 2.
C-09	Trust spec defines schema (step 1) before any evidence flows	🟢 Moderate	Evidence spine first, score second. Corrected in Part 2.
C-10	Doc type is operator-selected in some flows, auto-detected in others	🟢 Moderate	Auto-detect always; operator corrects. ADR-009.
C-11	Network Reputation weights peer feedback by peer Trust — circular	🟡 High	Defer to v2, hold weight at 0, renormalize. ADR-006.
C-12	Events "are emitted" with no stated delivery guarantee	🟡 High	Transactional outbox + idempotency keys. ADR-007.

1.3 The gap that matters most

The Trust Score measures how confidently we read a document. It does not measure whether the document is true.

Nothing in either spec checks authenticity. A fabricated SCA lab report, cleanly typeset and photographed well, will produce *higher* field confidence than a genuine crumpled one — and therefore earn *more* Trust. The system as specified rewards document quality, not honesty, which inverts the entire brand promise.

This is fixable, and the fix is already sitting in your own prior work:

1. **Issuer verification.** Third-party-issued documents (VCCI Certificate of Origin, plant protection department phytosanitary, a registered Q lab) carry a verifiable issuer. Self-issued documents (an exporter's own cupping form, their own invoice) do not. Weight them differently — a self-issued document should contribute a fraction of what an independently issued one does. The Trust spec gestures at this ("higher weight for multi-party cross-checks") but never operationalizes it.
2. **Cross-document consistency.** A CoO, a phytosanitary certificate, and a packing list for the same shipment share exporter, consignee, and quantity. Agreement across independently issued documents is a real authenticity signal. Disagreement is a red flag worth more than any single high-confidence extraction.
3. **Physical ground truth.** You already have a retained-sample procedure and a cupping SOP with calibration tolerances. That is the anchor. Quality Consistency should be scored against *your own re-cup of the retained sample*, not merely against the supplier's claim versus the supplier's lab report. Two documents that agree with each other prove nothing if both came from the same party.

Recommendation: add an `issuerClass` field (`self_issued | counterparty_issued | independent_third_party | government`) to every document type, and make it a multiplier on the Document Verification sub-score. This is a small schema change now and an expensive retrofit later.

1.4 Gap register — items neither spec covers

ID	Gap	Why it matters	Sprint
G-01	FX rate capture on USD-contract / VND-settlement pairs	Transaction Integrity will score correct payments as underpayments without a recorded rate and source	3
G-02	No appeal path for a negative Trust delta	A score that gates money with no correction route is a legal and relationship hazard	5
G-03	Automated decision-making / data protection	Smallholders are often natural persons. GDPR Art. 22 and Vietnam's Decree	0 (assess), 5 (implement)

ID	Gap	Why it matters	Sprint
		13/2023 (PDPD) plausibly attach. Get counsel — this is not a question to answer in-house	
G-04	Trust gaming: document farming, sybil parties, reciprocal feedback	Sealed band unlocks fast-path settlement. Anything that unlocks money gets attacked	6
G-05	`modelVersion` migration policy	Snapshots preserve the version, but nothing says what happens to displayed scores when weights change	4
G-06	Document deduplication	The same contract PDF uploaded by both parties should be one piece of evidence, not two	4
G-07	Document authenticity (see 1.3)	The central hole	1 (schema), 5 (scoring)
G-08	Bundled PDFs (contract + CoO + phyto in one file)	Extremely common in trade. Single-type-per-file breaks on real mail	3
G-09	Vietnamese-languag e handling: diacritics, red chops over text, mixed VN/EN	Determines whether accuracy targets are reachable at all	1
G-10	Operator calibration	Two operators accepting different things makes Trust noise, not signal	5 (SOP in Appendix A)

ID	Gap	Why it matters	Sprint
G-11	Per-document cost model (vision tokens × pages × reprocesses)	OCR unit cost against lot margin decides whether reprocess is free or rationed	1
G-12	Offline capture at origin and port	Field use is a stated requirement; no offline queue is specified	6 or v2
G-13	Deletion vs. evidence	7-year retention with soft-delete — does deleting a document reverse its Trust contribution?	4
G-14	Admin override design	Listed in success metrics, never designed	5
G-15	Golden set doesn't exist	The ">90% at ≥0.85 confidence" target is unmeasurable until 20+ real documents are collected and labeled	0 (start collecting)

G-15 has the longest lead time and zero engineering dependency. Start it in Sprint 0 and let it run in the background — collecting twenty real Vietnamese coffee documents with correct expected output takes weeks of asking people for files, not hours of work.

1.5 Architecture Decision Records

ADR-001: Trust Score scope — platform-global or tenant-local

Status: Proposed · **Deciders:** Founder + architecture

Context. The `documents` table is tenant-scoped. The Trust spec says Trust "attaches permanently to every tier profile" and "travels with the account." Those are incompatible without a decision. If Roaster A can see a Trust Score partly built from Roaster B's private contracts,

you are leaking commercial relationships. If Trust is purely tenant-local, it is worthless to a buyer evaluating a new supplier — which is the entire point.

Options.

	A: Tenant-local only	**B: Platform-global**	**C: Two-layer (recommended)**
Complexity	Low	Medium	Medium-High
Buyer value	Near zero for new suppliers	High	High
Leakage risk	None	Serious	Controlled
Cold start	Every tenant restarts at 50	Solved	Solved

Decision. Option C. A platform-level `parties` registry holds the canonical counterparty identity. Platform Trust is computed only from evidence the counterparty has consented to contribute (consent captured at onboarding, revocable). Each tenant additionally sees a private view incorporating their own documents. The public number is always the platform number; the private delta is labeled as such.

Consequences. Easier: cross-tenant supplier evaluation, the whole Navigator ranking story. Harder: party identity resolution and merge (the same exporter under three spellings), consent UX, and a `parties` table that must exist in Sprint 1 rather than Sprint 4. Revisit when the party graph exceeds a few thousand nodes.

ADR-002: OCR pipeline architecture

Status: Proposed · **Deciders:** Founder + engineering

Context. Two specs describe two pipelines. One says port a multi-engine orchestrator (cloud vision + Tesseract + layout model) feeding a separate LLM extraction step. The other says `services/docintake` already does this with Claude vision and structured outputs in a single pass.

Analysis. Multi-engine has a real cost: two systems to tune, two failure modes, two cost lines, and a fusion layer that itself needs confidence logic. For photographed Vietnamese documents specifically, Tesseract's `vie` model degrades badly on skew, diacritics, and the red official chops

that overlay text on phytosanitary certificates and CoOs. It would add work and subtract accuracy on exactly the document classes that matter most.

But blanket single-engine has its own hole: a vision-LLM's self-reported confidence on a number is not independent evidence that the number is right. For a cup score or a unit price — fields that move money — you want a second opinion from a system that fails differently.

Decision. Single vision-LLM as the primary path (whatever `docintake` already runs, pending the Sprint 0 audit). Add a **two-source rule** on critical numeric fields only: cup score, moisture percent, volume, unit price, total. Those get a second extraction pass — a cheap targeted re-read, or a deterministic OCR crop of the field region — and agreement between sources becomes part of the field confidence. Disagreement forces review regardless of either source's self-reported confidence.

Consequences. Easier: one pipeline to operate, much lower latency toward the <45s target, and a genuinely stronger signal where it counts. Harder: the two-source logic is new code with no reference implementation in either repo. Revisit if extraction accuracy on the golden set plateaus below target.

ADR-003: Money representation

Status: Proposed

Context. The Scanner spec says convert all money to integer cents before persistence. VND has no minor unit in practice; domestic coffee trades around 85,000–87,000 VND/kg. Storing that as "cents" multiplies every value by 100 for no reason and invites off-by-100 bugs at every boundary.

Decision. `Money { amountMinor: bigint, currency: ISO4217, scale: number }` with per-currency scale (USD = 2, VND = 0). Never a float, never a bare number. Additionally, every cross-currency record stores `fxRate`, `fxRateSource`, and `fxRateAt` — because contracts are quoted in USD and settled in VND via VietQR, and Transaction Integrity scoring compares those two numbers.

Consequences. Easier: correct arithmetic, and payment matching that doesn't produce phantom underpayments. Harder: every money field in the Zod contracts needs this shape from day one. This is cheap now and near-impossible to retrofit after there is production data.

ADR-004: Confidence policy location and reconciliation

Status: Proposed

Decision. One module, `contracts/confidence-policy.ts`, versioned alongside the extraction model. It exports the field-level bands (green ≥0.85, amber 0.70–0.84, red <0.70 — the Trust spec's scheme, since that is what the UI renders) and separately the document-level gate (≥0.92 overall with no critical field below 0.85 → auto-suggest accept). The two schemes are not in conflict once it is stated that one governs badges and the other governs the accept affordance. Both the worker and the UI import from this module. A CI check fails the build on any hardcoded threshold elsewhere.

ADR-005: Trust score storage and recomputation

Status: Proposed

Decision. Append-only `trust_score_snapshots` in Postgres, one row per score change, carrying components, `modelVersion`, and contributing `evidenceIds`. No Timescale in v1 — introducing a second database technology for one trend line on one panel is architecture debt purchased for a sparkline. Recomputation is a deterministic pure function of the evidence log plus a model version, so any historical score can be rebuilt rather than merely retrieved. Revisit at roughly a million snapshots.

ADR-006: Network Reputation component

Status: Proposed

Context. The spec gives Network Reputation 8% weight, defined as peer feedback weighted by the peers' own Trust. That is a recursive definition — A's score depends on B's, which depends on A's. It needs iteration to a fixed point with damping, collusion detection for reciprocal-feedback rings, and a defense against a party creating peers. It also cannot function before a party graph exists.

Decision. Hold Network Reputation at 0% for v1 and renormalize the remaining four components to 100% (Document Verification 38%, Transaction Integrity 27%, Quality Consistency 22%, Identity & Longevity 13%). Ship the component as a declared, visible zero — the panel shows it greyed with "not yet active" — so the model's shape is honest and v2 does not require a weight migration that shifts every existing score.

Consequences. Easier: no eigenvector math, no collusion detection, in v1. Harder: the "network" story is unavailable in early sales conversations. Revisit once there are enough completed cross-party transactions to make peer feedback meaningful.

ADR-007: Event delivery guarantees

Status: Proposed

Decision. Transactional outbox. Domain events are written to an `outbox` table in the same transaction as the state change, and a relay publishes them. Every consumer is idempotent on `(eventId)`. Every accept carries a client-supplied idempotency key. Without this, "accepting the same documentId twice returns the existing domain record" is a hope rather than a guarantee, and Trust — which is a running total driven by events — will drift silently. Trust totals that drift are worse than no Trust at all, because they are wrong and confident.

ADR-008: Greensheet → Auctum rename

Status: Proposed

Decision. Execute the rename in Sprint 0, at the package, table-prefix, and event-namespace level, before there is production data or an external integration. The cost curve on this is brutally steep — it is an afternoon now and a multi-week migration with a compatibility shim in nine months. If the decision is instead to defer, write down the boundary explicitly: which names are frozen legacy and which are current, so new code doesn't add to the split.

ADR-009: Document type classification

Status: Proposed

Decision. Always auto-classify; the operator corrects if wrong. Never require an upfront type selection. Some Trust-spec entry points ("Scan lab report / contract") imply a pre-choice — those become context hints that bias the classifier, not gates on the operator. Rationale: forcing a choice adds a decision to every capture (Hick's law tax on the highest-frequency action), and it is a choice the model makes better than a field operator holding a phone at a warehouse door.

1.6 Tech debt register

Scored as **(Impact + Risk) × (6 − Effort)**, per the standard framework. Higher is more urgent.

Item	Type	Impact	Risk	Effort	Score	Sprint
Live API keys in shared documents	Dependency/security	3	5	1	**40**	0
No transactional outbox / idempotency keys	Architecture	4	5	2	**36**	1
Money-as-cents assumption vs VND	Code	4	4	2	**32**	1
Two contradictory OCR architectures documented	Documentation	4	4	2	**32**	0
No golden document set	Test	5	3	3	**24**	0→3
`greensheet` naming throughout an "Auctum" product	Code	3	3	2	**24**	0
Trust weights hardcoded rather than versioned config	Architecture	3	3	2	**24**	4

Item	Type	Impact	Risk	Effort	Score	Sprint
No `issuerClass` on documents	Architecture	4	4	3	**24**	1
Thresholds duplicated across worker and UI	Code	3	3	2	**24**	1
Duplicate spec documents in Drive	Documentation	2	2	1	**20**	0
No visual regression on Trust bands	Test	2	3	2	**20**	6
No per-document cost instrumentation	Infrastructure	3	2	2	**20**	1
Design tokens not enforced in CI	Infrastructure	2	2	2	**16**	6

The top four are all Sprint 0 or 1. None of them are features, and that is the point: the sequencing error in both source specs is that they open with schema definition, when the actual first-order risks are secrets, delivery guarantees, and an unresolved question about what already exists.

1.7 Design system audit and critique

Against the Museum Folio system as described in the Trust spec.

Critique: Trust band encoding

Finding — colour-only band encoding. 🔴 Critical. The spec says "Never show a score without its band color." That is the wrong invariant. Colour is the least reliable channel: roughly 1 in 12 men have a colour vision deficiency, and your five-band palette (brass, sage, oxblood, neutral, danger) contains two red-adjacent tones — oxblood for Established and danger for At Risk — that are the exact pair most likely to collapse for a deuteranope. Someone would read "Established" as "At Risk" and hold a payment.

Recommendation. Invert the rule: *never show a band colour without its label.* The badge carries the numeral plus the band word, always. Colour is reinforcement, never the carrier.

Finding — semantic collision with cup score. 🟡 Moderate. Sealed Trust uses "brass-300 bg + ink text + seal glyph," described in the spec as "earned brass, same language as Outstanding cup score." Two different meanings now share one visual code. On a lot card that shows both, a buyer will read brass as "good coffee" and brass as "trustworthy party" without distinguishing which claim is which — and those are precisely the two things your brand exists to separate.

Recommendation. Keep the brass, change the shape language. Cup score stays a filled circular badge; Trust becomes a shield or seal outline with a hairline. Never render them adjacent without labels. Consider testing whether brass for Trust is worth the confusion at all.

Finding — contrast unverified. 🟡 Moderate. `oxblood-100 tint + oxblood-700 text` for Established, and `brass-300 bg + ink-900` for Sealed, both need measured contrast ratios. Tint-on-tint at 100/700 within one hue frequently lands under 4.5:1. The badge is small mono text — it needs the AA threshold, not an approximation of it.

Recommendation. Measure all five band combinations against WCAG 2.2 AA before the atom ships. Add the check to the token CI job.

Finding — metric row overload. 🟡 Moderate. The lot card already carries Cup Score, ESG, and four metric bars (Cost / Quality / ESG / Logistics). The spec proposes adding a Trust badge *and* an optional fifth bar. Six parallel bars is past the point where anyone reads any of them; it becomes texture.

Recommendation. Badge only on the card. Bars only in the Trust panel, where there is room and intent. If a fifth bar is added, remove one.

Finding — At Risk in public catalog. 🔴 Critical, and partly legal. The spec says At Risk parties get "limited visibility in public catalog" while the badge "may be visible" to unauthenticated users. Publicly rendering a named business with a danger-tinted "At Risk" label

is a defamation surface, and under ADR-001's global model it would be visible to that party's other customers.

Recommendation. At Risk and Provisional are internal bands. Publicly, a party either shows a badge (Established and above) or shows nothing. Absence communicates enough without making an assertion you would have to defend.

Audit: component completeness

Component	Variants	States specified	Docs	Score
TrustBadge	✅ 5 bands, 2 sizes	⚠️ no loading, stale, or empty	⚠️ spec only	6/10
Trust panel	✅	❌ no empty, loading, error, recomputing	❌	4/10
Capture surface	✅ desktop + mobile	⚠️ idle and active only	⚠️	6/10
Review pane	✅	⚠️ no partial-page-failure or unsupported-language state	⚠️	6/10
ConfidenceBadge	✅ 3	✅	⚠️	8/10

Missing states across the board. Every one of these components needs, and none currently specify: empty (zero documents — the spec handles this only for the lot card's "Add evidence" link), loading skeleton, error, and **stale** — a score computed under a superseded `modelVersion`, which will happen the first time weights change and needs a visual answer before it happens rather than after.

Token coverage. The spec's discipline here is genuinely good — "no arbitrary hex," everything referencing Museum Folio tokens, with a CI token check listed in success metrics. Keep that. Add the two-colour-scale caution: the Review pane renders confidence badges (green/amber/red status tokens) while the profile renders Trust bands (brass/sage/oxblood/neutral/danger). Those are two semantic scales that will appear within one workflow. Verify they read as distinct families and not as one confusing seven-colour system.

What works well

- Grounding the visual language in the archival/folio metaphor gives the product a coherent voice that a generic dashboard palette would not.
- JetBrains Mono tabular-nums for all numerics — scores, prices, confidence — is exactly right for scanning columns of figures.
- "Trust never silently gates access without explanation" is the strongest single line in either document. It should be a tested invariant, not a design aspiration.
- Explicitly naming what is *not* a scanner entry point (marketing surfaces, COF flows) is unusually disciplined. Most specs only say what to build.

1.8 Process optimization — the review bottleneck

Human review is the throughput ceiling of this entire system, by design. That is correct for correctness and dangerous for scale. If a lot carries six documents and an operator spends three minutes on each, a hundred lots is thirty operator-hours.

Current state (as specified)

Capture → upload → wait (OCR ~45s) → operator opens review

→ reads every field → corrects low-confidence → matches lot

→ accepts → next document (context switch back to capture)

Waste identified

Type	**Where**	**Cost**
Over-processing	Operator reviews every field, including high-confidence ones	Most of the review time on most documents
Manual work	Doc type selected by hand at several entry points	A decision per capture, on the highest-frequency action
Rework	Reprocess re-presents the full form; operator re-reads unchanged fields	Full review cost on every reprocess
Over-processing	Fields already known from the linked lot are re-verified from scratch	Exporter, consignee, quantity re-entered per document

Type	Where	Cost
Waiting	Operator waits ~45s at the capture surface for OCR	Idle time, and a context switch if they leave
Handoff	Field capture and desk review are the same session	Forces review in bad conditions (phone, warehouse, port)

Future state

1. **Auto-classify, operator corrects** (ADR-009) — removes a decision from every capture.
2. **Confidence-sorted queue** — clean documents first. Operators clear a run of easy ones fast and build calibration before hitting the hard ones.
3. **"Accept all green" bulk action** — one keystroke accepts every field above threshold; review collapses to only the amber and red fields. This is the single largest reduction available.
4. **Cross-document prefill** — a CoO, phytosanitary certificate, and packing list for one shipment share exporter, consignee, and quantity. Prefill from already-accepted documents on the same lot, and flag *disagreement* rather than asking for re-entry. This doubles as the authenticity cross-check from §1.3 — one mechanism serving two purposes.
5. **Diff-only reprocess** — after a reprocess, show only fields whose values changed.
6. **Decouple capture from review** — capture at origin or port on mobile, review at a desk. Nobody should be reading a nine-field contract form on a phone in a warehouse. Mobile gets capture, queue status, and a single-field correction affordance; full review is desktop.
7. **Queue-based, not modal** — the operator works a list, not a series of pop-ups tied to individual uploads.

Modelled impact

These are estimates from the waste analysis, not measurements. Instrument the current path in Sprint 2 and replace them with real numbers.

Measure	Now (est.)	After (est.)	Basis
Clean SCA report review	~90s (spec target)	~25–35s	"Accept all green" removes per-field reading on the majority of fields

Measure	Now (est.)	After (est.)	Basis
Contract review	~4 min (est.)	~2 min	Counterparty prefill and lot matching pre-resolved
Second document on the same lot	Full cost	~40% of full	Cross-document prefill
Reprocess	Full cost	~20% of full	Diff-only
Captures requiring a type decision	100%	~10% (corrections)	Auto-classification

The measurement that matters most is not median time. It is **operator correction rate on critical fields**, which the Scanner spec already targets at <15%. That number is the honest signal about whether extraction is good enough to lean on, and it should be on a dashboard from Sprint 2 onward — not computed at the end.

1.9 Risk register

Risk	Likelihood	Impact	Mitigation	Owner
Forged or altered documents inflate Trust	High	Severe — inverts the brand promise	`issuerClass` weighting, cross-document consistency, retained-sample ground truth (§1.3)	Architecture
Trust gaming via low-value document farming	High	High	Diminishing returns per type, distinct-issuer requirement, minimum completed transactions before Verified+	Architecture
Vietnamese-language extraction underperforms	Medium	High — blocks the accuracy target	Golden set from Sprint 0; measure before	Engineering

Risk	Likelihood	Impact	Mitigation	Owner
			committing to targets	
Automated-decision regulatory exposure (GDPR Art. 22, Vietnam PDPD)	Medium	Severe	Legal review in Sprint 0; human override + explanation + appeal by Sprint 5	Founder + counsel
Reputational/defamation exposure from public negative bands	Medium	High	At Risk and Provisional internal-only (§1.7)	Design + counsel
OCR unit cost exceeds per-lot margin	Medium	Medium	Cost instrumentation from Sprint 1; reprocess rationing if needed	Engineering
Operator inconsistency makes Trust noisy	High	Medium	Review SOP + calibration (Appendix A)	Operations
Scope creep from the Omni-Expert document	High	Medium	That document is a knowledge-base spec, not a sprint input. Explicitly out of scope — see §2.7	Founder

On that last row: the Coffee Industry Omni-Expert document describes an enormous capability surface — agronomy planning, futures forecasting, curriculum design, regulatory navigation. None of it belongs in these seven sprints. It is a roadmap for a different product. Keeping it visibly out of scope is itself a planning decision worth writing down.

One further note on that document. It repeatedly proposes "strategic circumvention," "bonded warehousing loopholes," "transshipment pathways," and "tariff engineering" as a headline capability. Tariff engineering and FTA optimization are legitimate and widely practiced. Transshipment structured to obscure origin, and HS reclassification that misstates the goods, are not — they are customs fraud, and under EUDR they are a criminal enforcement matter rather than a commercial one. For a platform whose entire proposition is verified provenance, marketing circumvention capability alongside a Trust Score is a positioning contradiction a competitor or a journalist would find in an afternoon. Worth a hard look at that framing before any of it becomes customer-facing copy.

PART 2 — FINAL SPRINT PLAN

2.1 Assumptions

State these as assumptions because they change the calendar, not the sequence.

- **Sprint length:** 2 weeks.
- **Team:** effectively one senior full-stack engineer plus AI-assisted execution, with design and operations handled by the founder against the expert-role framework. Sprints are sized as scope units. With one engineer, expect 14–16 weeks end to end. With two engineers running the OCR-worker and Review-UI streams in parallel from Sprint 2, expect 9–11 weeks.
- **Stack:** existing Auctum/Greensheet monorepo, tRPC, Postgres, Zod contracts, S3-compatible storage, event-driven with domain events. No new database technology introduced in v1.
- `services/docintake` **status:** unknown. Sprint 0 resolves this, and Sprint 1's scope adjusts by roughly one sprint depending on the answer.

2.2 Sprint map

Sprint	Theme	Ships	Gate to next
0	Truth & foundations (1 week)	ADRs signed, docintake audit, secrets rotated, golden-set collection started	All nine ADRs decided

Sprint	Theme	Ships	Gate to next
1	Evidence spine	Schemas, tables, upload→OCR→persist, outbox events	A photographed lab report becomes a reproducible `ocr_results` row
2	Review pane + QC accept	Side-by-side review, auto-classify, lot match, `qc_audits` creation	Photo → QC audit record, end to end, on the golden set
3	Contracts, money, AR	Contract extraction, counterparty match, Money+FX, bundled PDFs	A signed contract PDF → `contracts` row with correct VND/USD
4	Trust v1 (evidence-only)	Trust tables, calculator, events, badge behind flag	Accepting a document moves a Trust number, deterministically
5	Trust v1.0 + surfaces	Full model, panel, badges live, appeal + override	Every gate explains itself; appeals are recorded
6	Policy gates & hardening	Settlement gates, Navigator signal, anti-gaming, retention, regression	Golden-set regression green; token CI green

First visible win: end of Sprint 2 — a real photograph of an SCA lab report becomes a QC audit record with the operator in the loop. That is the demo that makes everything else fundable and every subsequent sprint concrete.

Sprint 0 — Truth & Foundations *(1 week)*

Goal: stop building on assumptions. Nothing here is a feature.

#	Task	Owner role	Done when
0.1	Rotate and revoke all exposed API keys; audit provider logs; scrub documents	Founder	Zero live credentials in any repo or Drive document
0.2	Audit `services/docintake`: what exists, what it extracts, what confidence it reports, what it costs per document	Engineering	Written gap doc against the Scanner spec
0.3	Decide ADR-001 through ADR-009	Founder + architecture	All nine marked Accepted with a date
0.4	Execute the `greensheet` → `auctum` rename, or write down the frozen boundary	Engineering	Either done or documented
0.5	Begin golden-set collection: 20+ real Vietnamese coffee documents across all eight types	Founder / operations	≥10 collected; target 20 by Sprint 3
0.6	Legal scoping call: automated decision-making, GDPR Art. 22, Vietnam Decree 13/2023, defamation exposure on published scores	Founder + counsel	Written position; constraints fed into Sprint 5
0.7	Consolidate the duplicate spec files; move the canonical version into `docs/` in-repo	Documentation	One file, versioned with the code

#	Task	Owner role	Done when
0.8	Stand up the task board with these sprints and owners	Founder	Backlog visible with owners and dates

Exit criteria: every ADR decided · docintake reality documented · no live secrets · golden set underway.

Sprint 1 — Evidence Spine

Goal: a document goes in and comes out as structured, confidence-scored, reproducible data. No review UI beyond a bare list.

Scope

- Zod schemas for all eight `documentType` payloads in `contracts/`, each including `issuerClass` (G-07) and `Money` per ADR-003
- `contracts/confidence-policy.ts` — the single versioned threshold module (ADR-004)
- Tables and migrations: `parties` (ADR-001), `documents`, `ocr_results`, `outbox`
- `documents.upload` mutation: auth, MIME and size validation, S3 write, row creation, job enqueue, immediate `documentId` return
- OCR worker: pre-process → extract → persist `ocr_results` with per-field confidence, `engineVersions`, `processingMs`
- Two-source rule on critical numeric fields (ADR-002)
- `document.ocr.completed` and `document.ocr.failed` through the outbox, with idempotency keys (ADR-007)
- Retry with exponential backoff, max 3, then `ocr_failed`
- Cost instrumentation: tokens and pages per document, logged per run (G-11)
- Vietnamese-language spike: run whatever golden-set documents exist through the pipeline and record accuracy honestly, including on chop-overlaid certificates (G-09)

Explicitly not in this sprint: any Trust logic, any review UI beyond a status list, any domain-record creation.

Exit criteria: a photographed SCA lab report produces an `ocr_results` row with per-field confidence · running the same document twice produces one document row and no duplicate events · the Vietnamese accuracy spike has a number attached to it.

Risks: if the Sprint 0 audit finds docintake is materially less complete than the Trust spec implies, this sprint expands. Decide at the Sprint 0 exit whether to split it into 1a and 1b rather than letting it silently overrun.

Sprint 2 — Review Pane and QC Accept

Goal: the first end-to-end business outcome. This is the demo.

Scope

- Review pane, Museum Folio, per the handoff spec in Appendix B: side-by-side original with zoom/pan, editable fields, confidence badges
- Auto-classification with operator correction (ADR-009)
- Lot matching: exact on `lotCode` and `sampleId`, fuzzy fallback, shown as selectable chips, operator can override
- `documents.acceptOcr` → creates `qc_audits`, sets `linkedEntityType`/`linkedEntityId`, status → accepted, idempotent on the key
- `documents.rejectOcr` and `documents.reprocess`
- `qc.audit.completed` through the outbox
- Confidence-sorted review queue (§1.8 item 2)
- "Accept all green" bulk action (§1.8 item 3) — the highest-value single interaction in the product
- Keyboard-first path: Tab cycles fields, Enter accepts when critical fields clear threshold or have been touched
- Document list with status chips and filters; timeline on document detail
- **Instrumentation:** median review time and per-field correction rate, on a dashboard, from day one of this sprint

Exit criteria: photograph of a real lab report → `qc_audits` record, unassisted, in under 90 seconds median across the golden set · double-accept creates one record · correction rate on critical fields is a number on a screen.

Sprint 3 — Contracts, Money, and AR

Goal: the commercial half of intake.

Scope

- Contract extraction: parties, lot references, volume, price terms, currency, delivery window, payment schedule, Incoterms
- Counterparty fuzzy matching against `parties`, with an explicit "create new party" path and merge handling
- Money and FX end to end (ADR-003, G-01): quoted currency, settlement currency, rate, source, timestamp
- AR schedule pre-population on accept
- `contract.created_from_ocr` through the outbox
- Bundled-PDF handling (G-08): page-range splitting so one file can yield a contract plus a CoO plus a phytosanitary certificate as separate documents
- Cross-document prefill (§1.8 item 4) — shared fields propagate across documents on one lot, and disagreement is surfaced rather than silently overwritten
- Diff-only reprocess view (§1.8 item 5)
- Golden set completed to 20+ documents; regression suite runs against it

Exit criteria: a real signed contract PDF → `contracts` row with correct currency handling and a populated AR schedule · a bundled three-document PDF splits correctly · golden-set regression passes.

Sprint 4 — Trust v1, Evidence-Only

Goal: Trust moves, deterministically, invisibly. No policy effects, no customer-visible badges.

Scope

- `trust_scores`, `trust_evidence`, `trust_score_snapshots` (ADR-005)
- Trust calculator as a pure function of the evidence log plus `modelVersion`
- v1 components live: **Document Verification (38%)** and **Identity & Longevity (13%)** only, normalized. Transaction Integrity and Quality Consistency are declared and inactive. Network Reputation is declared at zero (ADR-006)
- `issuerClass` multiplier on Document Verification (§1.3)
- Document deduplication (G-06): the same file uploaded by two parties is one piece of evidence
- Deletion policy (G-13): a soft-deleted document's Trust contribution reverses on the next recompute, and the reversal is itself an audited event
- `modelVersion` migration policy (G-05): recompute-on-read for the current version, snapshots preserved for history, and a "stale" visual state defined for scores computed under superseded weights
- `trust.evidence_recorded` and `trust.score_updated` through the outbox
- `TrustBadge` atom shipped to the design system behind a feature flag, internal-only

Exit criteria: accepting a document produces a Trust delta · accepting the same document twice does not · a full recompute from the evidence log reproduces the stored score exactly · no user outside the team sees a Trust number.

Sprint 5 — Trust v1.0 and Surfaces

Goal: the full model, visible, with the accountability machinery attached before the policy gates that make it consequential.

Scope

- **Transaction Integrity (27%)**: on-time settlement, VietQR / `providerTxnId` matching, AR aging, chargebacks — using the ADR-003 FX record so cross-currency settlements score correctly
- **Quality Consistency (22%)**: anchored on retained-sample re-cup against the existing cupping SOP tolerance bands, with claimed-vs-lab as a secondary signal (§1.3)
- Trust panel organism: numeral, band label, component bars, evidence list, trend from snapshots
- Badge on lot cards (badge only, no fifth bar — §1.7) and profile headers
- Scanner entry points: lot detail "Attach evidence," QC Audit primary capture, contract create "Scan first," counterparty profile "Upload verification documents," warehouse goods-in
- **Appeal path (G-02):** any negative delta is contestable. Contested evidence is flagged, the delta is suspended pending review, and the resolution is recorded. Build this in the same sprint the negative deltas go live, not after
- **Admin override (G-14):** explicit, reason-required, fully audited, visible in the evidence list as a distinct entry type
- **Explanation surface:** every score renders why — which evidence, which components, which version. "Trust never silently gates access without explanation" becomes a tested invariant
- All missing component states from §1.7: empty, loading, error, stale, recomputing
- WCAG 2.2 AA contrast verified on all five bands; band label always rendered alongside colour
- Operator review SOP published and operators calibrated (Appendix A, G-10)

Exit criteria: full four-component model live · every displayed score is explainable in the UI · an appeal can be filed, suspends its delta, and resolves with an audit trail · contrast verified · At Risk and Provisional are internal-only.

Sprint 6 — Policy Gates and Hardening

Goal: Trust starts affecting money. Everything that makes that safe ships in the same sprint.

Scope

- Fast-path settlement for Sealed and Verified when contract and lab report are both accepted
- Additional document gates and payment holds for Provisional and At Risk, each with an in-UI explanation of exactly what would clear it
- Navigator ranking signal: document density and supplier Trust as composite inputs alongside cup score and budget; "Needs verification" chip on low-density lots
- **Anti-gaming (G-04):** diminishing returns per document type, distinct-issuer requirement for Document Verification to exceed a ceiling, minimum completed transactions before any party reaches Verified or above, reciprocal-pattern detection logged for review
- Retention and lifecycle: configurable per document class, soft-delete plus storage lifecycle rules, PII access audit log on contracts
- Data residency position documented for Vietnamese operations; regional OCR endpoints where available
- Golden-set regression in CI; visual regression on Trust bands; design-token CI check enforcing zero arbitrary colours

Exit criteria: a gate fires and the UI explains it in plain language · anti-gaming rules have tests that attempt the attack · golden-set regression and token CI both green in the pipeline.

2.3 Definition of Ready / Definition of Done

Ready — a ticket may enter a sprint when: the relevant ADR is Accepted; its Zod contract exists or is part of the ticket; its success metric is stated and instrumented somewhere; and its failure modes are listed.

Done — a ticket leaves a sprint when: unit tests cover the schema and threshold logic; integration tests cover the event path; the action is idempotent and proven so by a test; no hardcoded thresholds or hex values (CI-enforced); the operator-facing surface has empty, loading, and error states; and any metric it claims to move is visible on a dashboard.

2.4 Metrics

Instrument from Sprint 1; do not compute at the end.

Metric	Target	Source	From
Upload → ready-for-review, single page	<45s median	Worker timing	S1
SCA reports at documentConfidence ≥0.85, first pass	>90%	Golden set	S3
Operator correction rate on critical fields	<15%	Review telemetry	S2
Median clean-report review time	<90s, trending to <35s	Review telemetry	S2
Financial or quality records created without explicit accept	0	Audit log	S2
Cost per document (tokens × pages)	Under per-lot margin threshold	Worker instrumentation	S1
Lots with ≥1 accepted verification document within 7 days	Set baseline in S5, then target	Trust evidence	S5
Trust ↔ later dispute rate correlation	Negative	Trust + disputes	S6+
Trust updates without an evidence event or audited override	0	Audit log	S4
Arbitrary colours in Trust/Scanner components	0	Token CI	S6

2.5 Deferred to v2 — explicitly

Network Reputation scoring · real-time video stream OCR · handwriting-only forms without printed anchors · fully autonomous financial postings · Timescale · public Trust badges for negative bands · offline field capture queue (G-12, promote to v1 if origin capture proves connectivity-blocked) · everything in the Coffee Industry Omni-Expert document.

APPENDIX A — Process Document: Document Review and Accept

Owner: Operations lead · **Review cadence:** Quarterly · **Ships:** Sprint 5

Purpose

Ensure that every document accepted into Auctum Ledger has been verified by a qualified operator to a consistent standard, so that Trust Score movements reflect evidence rather than operator variance.

Scope

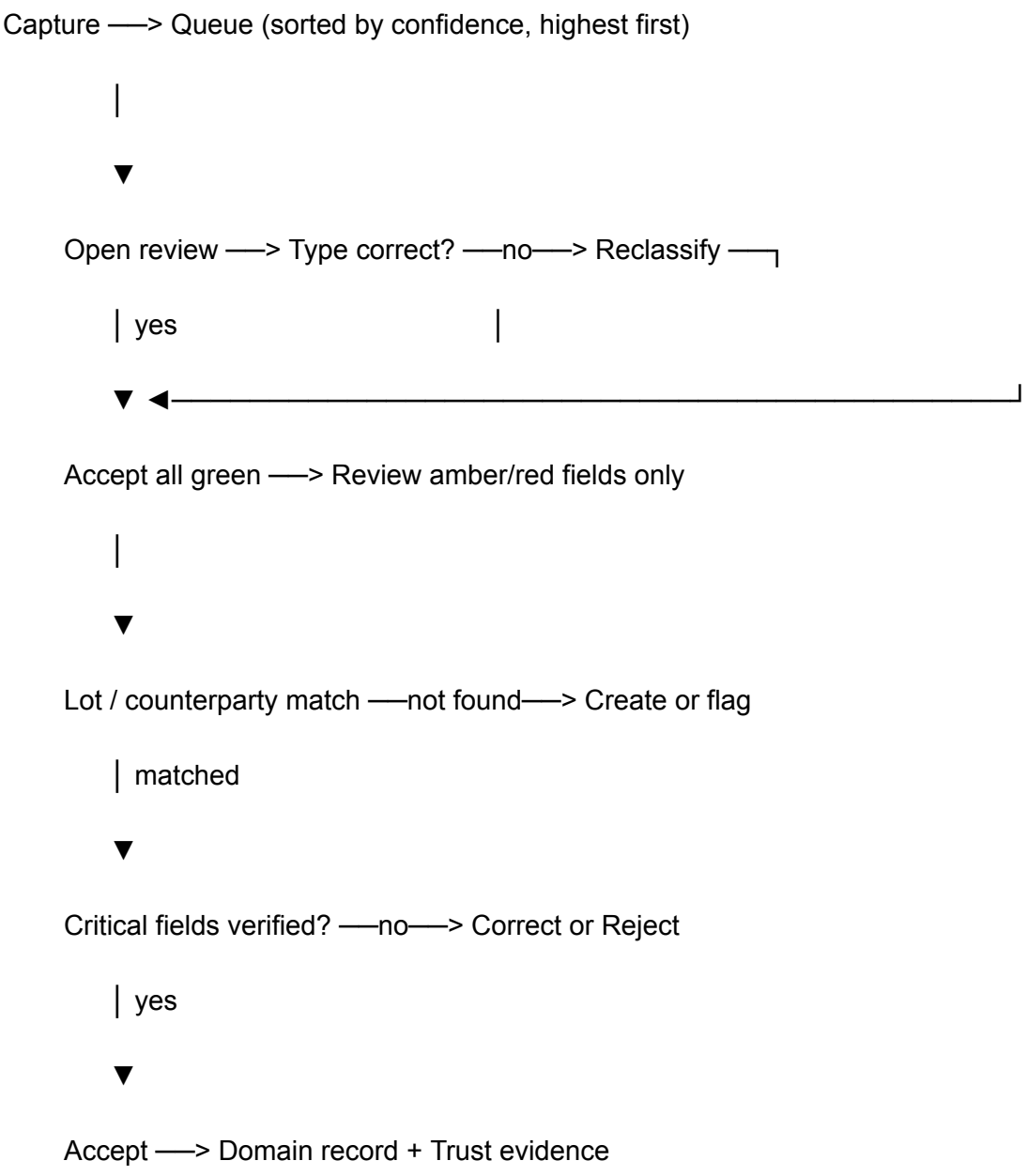
Covers review and accept/reject of all eight document types. Excludes physical QC (see the cupping SOP and retained-sample procedure) and payment authorization.

RACI

Step	Responsible	Accountable	Consulted	Informed
Capture	Field operator / QC staff	Ops lead	—	—
Classify correction	Reviewer	Ops lead	—	—
Field review	Reviewer	Ops lead	QC lead (quality fields)	—
Lot / counterparty match	Reviewer	Ops lead	Trade desk (contracts)	—
Accept	Reviewer (QC/Ops/Admin role)	Ops lead	—	Trust system
Reject	Reviewer	Ops lead	Uploader	Uploader

Step	Responsible	Accountable	Consulted	Informed
Appeal handling	Ops lead	Founder	Counterparty, counsel if escalated	Trust system
Admin override	Admin	Founder	Ops lead	Audit log

Flow



Detailed steps

Step 1 — Take from queue. Reviewer works the confidence-sorted queue top-down. Do not cherry-pick; skipping the hard ones makes the correction-rate metric lie.

Step 2 — Confirm classification. If the auto-detected type is wrong, correct it. A reclassification re-runs extraction against the correct prompt; expect a short wait.

Step 3 — Accept all green. One action clears every field above threshold. Only amber and red fields remain.

Step 4 — Verify critical fields against the image. For lab reports: `sampleId`, `lotCode`, `cupScore`, `moisturePercent`. For contracts: parties, volume, unit price, total, currency, payment terms. **Read these off the original image every time, regardless of confidence colour.** A confident wrong number is the failure mode this whole process exists to catch.

Step 5 — Resolve the match. Confirm the suggested lot or counterparty, or override. If no match exists, flag rather than force — a wrongly linked document is worse than an unlinked one.

Step 6 — Accept or reject. Reject with a reason when the document is illegible, the wrong document, or internally inconsistent. Rejection carries no Trust penalty for the uploader; that is deliberate, so nobody hesitates to upload a poor photo.

Exceptions

Scenario	Action
Extracted value contradicts existing lot data materially	Do not accept. Escalate to QC lead. This is a Quality Consistency signal, and it needs a human decision about which source is wrong
Bundled PDF with multiple document types	Split by page range, review each separately
Document in an unsupported language or script	Flag for manual entry; do not guess
Partial multi-page OCR failure	Review completed pages; flag incomplete ones; do not accept a partial contract
Counterparty disputes an accepted document	Open an appeal; the associated Trust delta is suspended pending resolution

Scenario	Action
Reviewer is unsure	Leave in queue and escalate. An unresolved document costs a day; a wrongly accepted one costs the score's credibility

Metrics

Metric	Target	Method
Correction rate on critical fields	<15%	Review telemetry
Median review time, clean lab report	<35s post-optimization	Review telemetry
Inter-operator agreement on a monthly calibration set	>90%	Quarterly calibration exercise, mirroring the cupping SOP's calibration model
Documents accepted then later appealed successfully	<2%	Appeal log

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APPENDIX B — Design Handoff: TrustBadge and Review Pane

Ships: TrustBadge in Sprint 4 (flagged) · Review pane in Sprint 2 · Trust panel in Sprint 5

B.1 TrustBadge (atom)

Purpose. A single legible signal of how much verified evidence stands behind a party or lot. Parallel in geometry to `CupScoreBadge`, deliberately distinct in shape.

Props

Prop	Type	Default	Notes
`score`	`number | null`	—	0–100. `null` renders the empty state, never a zero
`band`	`'sealed' | 'verified' | 'established' | 'provisional' | 'at_risk'`	derived	Derived from score; overridable only for storybook
`size`	`'sm' | 'md'`	`'sm'`	24px / 28px height
`showLabel`	`boolean`	`true`	**Cannot be set false in production.** Colour never carries the band alone
`modelVersion`	`string`	—	Drives the stale state
`evidenceCount`	`number`	—	Tooltip only

Bands

Band	Score	Background	Text	Glyph	Public?
Sealed	90–100	`brass-300`	`ink-900`	Seal outline	Yes
Verified	75–89.9	`sage-600`	white	—	Yes
Established	55–74.9	`oxblood-100`	`oxblood-700` + 1px hairline	—	Yes
Provisional	35–54.9	`neutral-700`	white	—	**Internal only**
At Risk	0–34.9	`danger` tint	`danger` text	—	**Internal only**

All five combinations require measured WCAG 2.2 AA verification before ship. Publicly, Provisional and At Risk render no badge at all rather than a negative one (§1.7).

States

State	Visual	Behavior
Default	Band background, mono numeral, band word	Tooltip on hover
Empty (no evidence)	No badge. `oxblood` text link "Add evidence"	Opens attach flow
Loading	Skeleton at badge dimensions	Non-interactive
Stale (superseded `modelVersion`)	Badge at 70% opacity + hairline dashed border	Tooltip: "Recalculating under model v[x]"
Recomputing	Skeleton with subtle pulse	Non-interactive
Error	Neutral badge, em-dash instead of numeral	Tooltip: "Score unavailable"

Typography and geometry

JetBrains Mono 700, `tabular-nums`. Height 24px (sm) / 28px (md). `radius-full`. Band label in the badge at `overline` treatment, never truncated — if space is short, drop the badge, not the word.

Accessibility

- **Role:** `img` with `aria-label="Trust Score 87, Verified, 12 accepted documents"`
- **Never** conveys band by colour alone
- Tooltip content must also be available to screen readers via the label, not only on hover
- Contrast: all five bands ≥4.5:1 for the numeral and label, verified by measurement

Do / Don't

✅ **Do**	❌ **Don't**
Always render the band word with the colour	Rely on colour to distinguish Established from At Risk
Use badge-only on lot cards	Add a fifth metric bar to an already-crowded row

✅ Do	❌ Don't
Use a distinct shape from `CupScoreBadge`	Reuse the cup-score brass treatment unchanged
Show "Add evidence" when there is none	Show a score of 0 or 50 for a party with no history

B.2 Review pane

Layout

Desktop two-column, roughly 55/45. Left: original page, paper-frame, hairline border, zoom and pan, page navigation for multi-page. Right: form fields on `surface`. Below 768px the layout stacks with the image collapsed to a thumbnail strip — and mobile is explicitly **not** the primary review surface (§1.8 item 6); it supports single-field correction, not full review.

Field row

Element	Spec
Label	`overline`, muted
Value	Editable input, JetBrains Mono for numerics, body face for text
Confidence badge	Right-aligned. Green ≥0.85, amber 0.70–0.84, red <0.70, from `confidence-policy`
Low-confidence treatment	`warning-bg` soft tint on the row; auto-focused in sequence
Two-source disagreement	Distinct treatment from low confidence — a hairline `oxblood` left border and an inline note naming both values. Do not overload the amber tint with a second meaning

Actions

Action	Variant	Behavior
Accept all green	Secondary, prominent	Clears every above-threshold field. Announce the count: "14 fields accepted"
Accept & Link to Lot / Profile	Primary (`oxblood`)	Enabled when critical fields clear threshold or have been touched
Reject	Outline	Requires a reason
Reprocess	Ghost	Diff-only view on return

States

State	Treatment
Loading OCR	Skeleton form with `oxblood` spinner; progress in mono. Operator can leave; the queue persists
Partial page failure	Completed pages reviewable; failed pages flagged inline with a per-page retry
Unsupported language	Clear message plus a manual-entry fallback path. Do not present a form of confident garbage
No lot match	Chips absent; "No matching lot" plus a create-or-flag choice. Never auto-create
Already accepted	Read-only with an "Accepted [date] by [user]" header

Keyboard

Tab cycles fields in confidence order, lowest first. `Cmd/Ctrl+Enter` triggers Accept when eligible. `Cmd/Ctrl+G` triggers Accept all green. Escape returns to queue with the draft preserved. Focus order must follow visual order; the auto-focus-lowest-confidence behavior operates on entry, not by reordering the DOM.

Motion

Field accept: 120ms ease-out fade on the badge. Page transitions in the image pane: none — instant. Zoom: 180ms ease-out. No parallax, no scroll-triggered reveal; this is a working surface, not a marketing page.

Immediate next actions

1. Rotate the exposed API keys. Today, before anything else.
2. Audit `services/docintake` and write down what it actually does. Every downstream estimate depends on the answer.
3. Decide the nine ADRs — particularly ADR-001 (Trust scope), which determines the Sprint 1 schema and cannot be deferred past it.
4. Start collecting the golden set. It has the longest lead time and blocks the accuracy targets in Sprint 3.
5. Get the legal scoping call on the calendar. Automated scoring that gates payment for parties who may be natural persons is not a question to resolve internally.